Microplastic contamination in agricultural soils: effects on crop nutrient profiles and implications for global food security policy

Abstract

Microplastic contamination of agricultural soils has emerged as one of the most significant yet underregulated threats to global food production. Agricultural soils now receive an estimated **110,000–730,000 tonnes of microplastics annually** in Europe and North America alone through sewage sludge application, ScienceDaily Phys.org plastic mulch films, irrigation water, and atmospheric deposition — a burden that likely exceeds the total mass of microplastics floating on the world's ocean surfaces. Recent meta-analyses demonstrate that microplastic exposure reduces plant photosynthesis by 7–12%, decreases crop biomass by approximately 13%, and diminishes key micronutrient concentrations in cereal grains by 10–17%. A landmark 2025 study published in the *Proceedings of the National Academy of Sciences* estimates annual global crop losses of **109–361 million metric tons** attributable to microplastic-induced photosynthetic impairment. PNAS Despite this mounting evidence, no country has established soil quality standards or maximum allowable concentrations for microplastics in agricultural soils. This paper synthesizes peer-reviewed evidence from 2018–2025, including recent meta-analyses, field studies, and policy analyses, to evaluate the extent to which microplastic contamination alters crop nutrient profiles and to assess the implications for food security regulation worldwide.

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1. Introduction

Agricultural soils have quietly become the world's largest reservoir of microplastic pollution. Terrestrial environments may contain **up to 20 times more plastic than oceanic systems**, Springer Frontiers yet research and regulatory attention have overwhelmingly focused on marine contamination (Büks and Kaupenjohann, 2020, *SOIL*, 6, 649–662, DOI: 10.5194/soil-6-649-2020). The disconnect between the scale of agricultural soil contamination and the near-total absence of regulatory response represents one of the most consequential policy gaps in contemporary environmental governance.

Microplastics — synthetic polymer particles smaller than 5 mm Frontiers — enter agricultural soils through multiple pathways: ACS Publications the application of sewage sludge and biosolids as fertilizer, the degradation of plastic mulch films, Frontiers irrigation with treated wastewater, atmospheric deposition, PubMed Central and contaminated compost and coated fertilizers. Frontiers +3 Once deposited, these particles persist for decades to centuries, accumulating with each successive application cycle and fragmenting into ever-smaller nanoplastics that can penetrate plant root and leaf tissues directly.

The implications extend beyond environmental contamination to fundamental questions about nutritional security. Emerging evidence demonstrates that microplastics alter soil structure, disrupt microbial communities essential for nutrient cycling, impair root function, and reduce the concentrations of critical macro- and micronutrients in harvested crops. Nature PubMed Central For the approximately 800 million people currently facing food insecurity, and the billions more dependent on soils receiving regular biosolid or plastic mulch inputs, these findings carry profound significance. A 2025 PNAS meta-analysis warns that microplastic-induced crop losses could increase the number of people at risk of starvation by approximately **400 million within two decades** EHN (Li, Zhong et al., 2025, *Proceedings of the National Academy of Sciences*, DOI: 10.1073/pnas.2423957122).

This paper examines four interconnected dimensions of this crisis: the sources, types, and geographic distribution of microplastic contamination in agricultural soils; the documented effects on crop nutrient profiles and plant health; the methodological challenges in detection and quantification; and the policy and food security implications demanding urgent attention.

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2. Sources, types, and accumulation of microplastics in agricultural soils

2.1 Sewage sludge delivers the largest microplastic loads to farmland

Biosolid application constitutes the single largest pathway for microplastic entry into agricultural soils across Europe, North America, and Australia. Approximately 50% of sewage sludge generated in Europe and North America is processed for agricultural use, ACS Publications and this material carries enormous microplastic loads. Nizzetto et al. (2016, *Environmental Science & Technology*, DOI: 10.1021/acs.est.6b04140) estimated that **110,000–730,000 tonnes of microplastics** transfer annually to European and North American farmland through sludge application ScienceDaily — equivalent to **125–850 tonnes per million inhabitants** per year. EHN

A systematic review of 65 studies across 25 countries found sewage sludge microplastic concentrations ranging from 0.193 to 1.69×10^5 particles per gram, with a **median of approximately 22,400 particles per kilogram** ScienceDirect (Harley-Nyang et al., 2023, *Science of the Total Environment*). Field studies have quantified the cumulative effect: in Valencia, Spain, each sludge application increased soil microplastic loads by approximately **710 particles per kilogram**, ScienceDirect with sludge-amended soils containing 5,190 particles/kg compared to 2,030 particles/kg in unamended soils ScienceDirect (Van den Berg et al., 2020, *Environmental Pollution*, 261, 114198). In Chile, median microplastic counts increased from 1.1 to 3.5 particles per gram across one to five sludge applications of 40 tonnes per hectare ScienceDirect (Corradini et al., 2019, *Science of the Total Environment*, 671, 411–420).

The persistence of sludge-derived microplastics is striking. A landmark 25-year temporal study published in 2025 documented that microplastic abundance increased **723–1,445%**

following sewage sludge applications and remained statistically constant over 22 years, showing no significant degradation of the bulk microplastic load (ScienceDirect) (*Chemosphere*, 2025, DOI: 10.1016/j.chemosphere.2025.144277). Similarly, Weber et al. (2022, *Scientific Reports*) found that 34 years after the last sewage sludge application in Germany, macroplastics persisted at 637 items per hectare on soil surfaces, with microplastics detected at depths of up to 90 cm. (Nature) In the United States, an estimated 21,249 metric tonnes of microplastics are released annually to agricultural lands through biosolid application alone. (EHN)

2.2 Plastic mulch film creates severe contamination in Chinese agriculture

China represents a distinct contamination paradigm, being the world's largest consumer of agricultural plastic mulch film, with approximately **2 million tonnes applied annually** across 18.4 million hectares — roughly 10% of the nation's cultivated land. Huang et al. (2020, *Environmental Pollution*, 260, 114096) sampled 384 sites across 19 Chinese provinces and found macroplastic concentrations averaging 83.6 kg/ha, (PubMed) with microplastic concentrations increasing dramatically over time: from **80.3 ± 49.3 particles/kg** after 5 years of continuous mulching to **1,075.6 ± 346.8 particles/kg** after 24 years. (Copernicus) A 32-year field trial in Shenyang demonstrated that (ScienceDirect) plastic film-derived microplastics contributed 33–56% of total microplastics throughout the 0–100 cm soil profile, with particles migrating to depths exceeding 80 cm. (ScienceDirect) (Nature)

Regional variation within China is substantial. Wuhan vegetable farmlands recorded concentrations up to **1.6×10^5 particles/kg**, while Yunnan planting areas ranged from 7,100 to 42,960 items/kg. (Nature) Greenhouse soils showed dramatically elevated levels, with averages of 2,110 particles/kg inside greenhouses compared to 310 particles/kg outside (Neptjournal) (Guizhou, China, 2024, *Scientific Reports*). The contamination problem is compounded by China's historical use of ultra-thin mulch films (ScienceDirect) (< 0.01 mm), which fragment readily and are nearly impossible to recover at end of season.

2.3 Irrigation, atmospheric deposition, and other pathways compound the problem

Beyond the two dominant pathways, several secondary sources contribute substantially to agricultural soil contamination. Irrigation with treated wastewater introduces **0.2–0.5 microplastic particles per milliliter**, representing as much as 55.9% of total microplastic input in irrigated orchards (PubMed Central) (Portuguese orchard study, 2025, *Environmental Monitoring and Assessment*). Atmospheric deposition delivers microplastics at rates of **19.4–406.9 particles per square meter per day**, accounting for over 40% of total inputs in some Mediterranean agricultural settings. (PubMed Central)

Compost and organic amendments carry significant loads: chicken manure has been found to contain **14,720–129,800 particles per kilogram**. (Wiley Online Library) Coated and slow-release fertilizers contribute an estimated **8,000–10,000 tonnes of plastic annually** to Western European agricultural soils, (Unep) with the European Chemicals Agency (2019) identifying

agriculture as the single largest sector for intentionally added microplastic use in the European Economic Area (EHN) — approximately **22,500 tonnes per year** in fertilizers alone. (Unep)

2.4 Global contamination patterns reveal polyethylene dominance and geographic disparities

A 2024 meta-analysis of 89 studies reported mean agricultural soil microplastic concentrations of **2,900 ± 7,600 particles/kg**, with maximum values for greenhouse-mulched soils reaching 5,200 ± 8,300 particles/kg and for sludge-amended soils reaching 3,700 ± 8,800 particles/kg (ScienceDirect) (*Science of the Total Environment*, 950, 175276). The first global review by Büks and Kaupenjohann (2020) examined 223 sites and found a median of 1,076 items/kg (interquartile range: 78–2,500), with municipal-vicinity soils showing concentrations approximately 10 times higher than rural agricultural soils. (Copernicus) (Scite)

Polyethylene (PE) is the most frequently detected polymer in agricultural soils, (Wiley Online Library) identified in over 50 studies globally, followed by polystyrene (PS, 37 studies) and polypropylene (PP, 32 studies). (ScienceDirect) (Wiley Online Library) Polyester (PET) fibers, polyvinyl chloride (PVC), polyamide, and polyurethane are also commonly detected. (Wiley Online Library) The dominant morphology varies by contamination pathway: **fibers constitute 90–97% of particles** in sludge-amended soils, (ScienceDirect) reflecting their textile origin in wastewater, while fragments and films dominate in mulch-contaminated soils. The majority of detected particles fall within the 10–1,000 μm size range, though analytical limitations mean smaller particles are substantially undercounted.

Geographic data coverage remains highly uneven. China and Europe dominate the published literature, (Copernicus) while Africa, Central Asia, and much of South America are severely underrepresented. (BIOENGINEER.ORG) A 2025 review identified only approximately four studies on microplastic monitoring in African terrestrial ecosystems (Springer) (Yh and Mc, 2025, *Microplastics and Nanoplastics*, 5, 15, DOI: 10.1186/s43591-025-00122-z). This data gap is particularly concerning given that developing nations with limited waste management infrastructure likely face substantial contamination from unmanaged plastic waste entering agricultural systems.

2.5 Microplastics persist for decades and migrate through soil profiles

Agricultural soils function as long-term accumulators rather than transient reservoirs. Field evidence consistently demonstrates persistence exceeding two decades with minimal natural loss. Schell et al. (2022) found that in semi-arid Central Spain, only **0.2–0.4% of soil microplastics** were mobilized by water runoff in a given year. (ScienceDirect) Conventional polyethylene is effectively non-biodegradable under soil conditions, persisting for centuries. (ScienceDirect) Even biodegradable plastics may require more than four years to reach 90% degradation under field conditions. (SpringerOpen)

Vertical migration proceeds slowly but steadily. A two-year Swiss field experiment (2025, *Communications Earth & Environment*) demonstrated that only approximately **1% of surface-applied microplastics** reached below 8 cm depth, indicating that microplastics accumulate primarily in topsoils, creating localized pollution hotspots within the root zone. Nature PubMed Central However, over longer timescales, bioturbation (by earthworms and soil fauna), agricultural tillage, wet-dry cycles, and irrigation facilitate deeper penetration. Frontiers The 32-year Shenyang trial documented microplastics at 100 cm depth, and Weber et al. (2022) found particles at depths up to 2 meters in floodplain soils through bioturbation and preferential flow pathways. Wiley Online Library

The fragmentation of existing macroplastics represents a continuously growing reserve pool. Copernicus UV photodegradation, thermal stress, and mechanical weathering during tillage progressively fragment larger plastic items into secondary microplastics and nanoplastics, ensuring that contamination intensifies even if all new inputs were eliminated.

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3. Effects on crop nutrient profiles, plant physiology, and food quality

3.1 Macronutrient uptake is disrupted through multiple pathways

Microplastic contamination alters the availability and plant uptake of nitrogen, phosphorus, and potassium through interconnected mechanisms affecting soil chemistry, microbial activity, and root function. The effects are polymer-type and concentration-dependent, with some low-dose scenarios producing hormetic (stimulatory) responses that reverse at higher concentrations.

Phosphorus availability appears most consistently impacted. Wang et al. (2024) found that soil available phosphorus content decreased by **9.7–38.6% under conventional microplastics** (PE, PVC) and by **38.4–73.6% under biodegradable microplastics** (PLA) across three soil types. PE and PLA microplastics significantly reduced soil urease activity by 15.32% and 40.38%, respectively, Frontiers impairing phosphorus cycling (2024, *Frontiers in Microbiology*, DOI: 10.3389/fmicb.2024.1513890). **Nitrogen** dynamics are altered in complex ways: microplastics increase the soil carbon-to-nitrogen ratio by **24–35%**, limiting nitrogen availability, Frontiers while simultaneously affecting nitrification, denitrification, and ammonia volatilization processes. Field studies have documented two-fold increases in N_2O emissions from maize agroecosystems with PE microplastic contamination. ScienceDirect **Potassium**, magnesium, and sulfur concentrations in onion shoots decreased under microplastic fiber exposure (Moreno-Jiménez et al., 2022, *Journal of Sustainable Agriculture and Environment*, 1, e12006), and all microplastic treatments containing PS, PE, and PET produced marked decreases in stem potassium content in red amaranth BMC Plant Biology (Roy et al., 2024, *BMC Plant Biology*, 24, 608, DOI: 10.1186/s12870-024-05312-0).

3.2 Micronutrient depletion in grains poses hidden hunger risks

The effects on essential micronutrients carry particular significance for nutritional security. In rice grains exposed to 250 mg/kg polystyrene nanoplastics (80 nm), **calcium content decreased by 16.33%, manganese by 16.86%, and zinc by 9.94%** compared to controls PubMed Central MDPI (Jadhav and Medyńska-Juraszek, 2024, *Plants*, 13, 2526, DOI: 10.3390/plants13172526). Lian et al. (2020, *Journal of Hazardous Materials*, 385, 121620) demonstrated that PS nanoplastics at 10 mg/L **inhibited the uptake and translocation of iron, manganese, copper, and zinc** in wheat seedlings despite initially promoting growth. In root vegetables, PET microplastics caused micronutrients to **increase in roots but decrease in shoots**, suggesting impaired root-to-shoot translocation Springer — a critical finding for crops whose edible portions are above ground (Shorobi and Park, 2025, *Applied Biological Chemistry*, 68, 66, DOI: 10.1186/s13765-025-01039-0).

PE microplastics specifically decreased manganese accumulation while increasing sodium in rice seedlings PubMed Springer (Tang et al., 2022, *Bulletin of Environmental Contamination and Toxicology*, 108, 949–955, DOI: 10.1007/s00128-021-03453-8). These reductions in micronutrient density, even when occurring alongside maintained or increased yields, mirror the phenomenon of "hidden hunger" — where food appears adequate in quantity but is deficient in essential micronutrients. PubMed Central MDPI Given that approximately **2 billion people** already suffer from micronutrient deficiencies globally, microplastic-driven nutrient dilution in staple cereals could significantly compound this challenge.

3.3 Soil structure and microbial disruption drive nutrient cycling breakdown

The mechanisms by which microplastics alter soil-plant nutrient dynamics operate at multiple scales. At the physical level, microplastics significantly modify soil properties: PubMed Central de Souza Machado et al. (2018, *Environmental Science & Technology*, ACS Publications 52, 6044, DOI: 10.1021/acs.est.8b02212) demonstrated that ACS Publications all four tested microplastic types — polyacrylic fibers, polyamide beads, polyester fibers, and PE fragments — **significantly altered soil bulk density** at concentrations up to 2%. ACS Publications +3 Fibers tend to increase water-holding capacity while reducing bulk density SCIRP by up to 15%, fundamentally changing the pore structure through which water, air, and nutrients move. PubMed Central Soil porosity decreased from **48% to 32%** in contaminated soils with increasing microplastic concentration, with polystyrene showing the strongest effect. PubMed Central

Microbial community disruption may represent the most consequential mechanism. Microplastics reduce soil enzyme activities critical for nutrient mineralization: Frontiers Nature phenol oxidase activity declined by **15.4–54.8%**, manganese peroxidase by **39.5–61.0%**, laccase by **16.5–57.6%**, and β-glucosidase by **30.6–42.7%** across various studies. Taylor & Francis Online Arbuscular mycorrhizal fungi (AMF), which facilitate phosphorus and micronutrient uptake for over 80% of terrestrial plant species, are particularly vulnerable. Springer Microplastics and cadmium together reduce AMF symbiosis, Maxa Press while biodegradable PLA microplastics

adversely affect AMF diversity and community structure. Frontiers Notably, AMF inoculation can buffer microplastic effects on nutrient uptake, particularly enhancing manganese uptake Wiley Online Library (Moreno-Jiménez et al., 2022), suggesting that maintaining soil biological health represents a critical defense against microplastic-driven nutrient depletion.

At the root level, microplastics adhere to root surfaces, creating **physical apoplastic barriers** that directly inhibit nutrient and water uptake Springer PubMed Central (Hussain et al., 2025, *NanoImpact*, 37, 100536). PE, LDPE, and PLA negatively affected root growth in maize at all growth stages. Furthermore, microplastics' large specific surface area Frontiers enables them to adsorb nutrients through competitive sorption, effectively sequestering available nitrogen and phosphorus from the soil solution.

3.4 Crop yield and nutritional quality decline at environmentally relevant concentrations

A comprehensive global meta-analysis integrating 3,286 data points estimated that microplastic exposure reduces terrestrial plant photosynthesis by **7.05–12.12%** and chlorophyll content by **10.96–12.84%**, translating to estimated annual global crop losses of **4.11–13.52%** or 109.73–360.87 million metric tonnes for wheat, rice, and maize PNAS (Li, Zhong et al., 2025, *PNAS*, DOI: 10.1073/pnas.2423957122). A separate meta-analysis of 128 studies found microplastics reduced **plant productivity by 14.5%**, soil fertility by 8.2%, and microbial biodiversity by 3.4% ScienceDirect (*Journal of Hazardous Materials*, 2024, DOI: 10.1016/j.jhazmat.2024.136379). An earlier meta-analysis documented that microplastics **inhibited fruit traits by 13.34%**, shoot traits by 9.7%, and root traits by 6.65%, while increasing oxidative stress markers — catalase (CAT) activity increased by 12.71% and malondialdehyde (MDA) content by 14.82% Wiley Online Library (H. et al., 2024, *Land Degradation & Development*, 35, 551–567, DOI: 10.1002/ldr.4957).

Specific nutritional quality impacts are well documented. In rice, PS microplastics inhibited **29.63% of substance accumulation-related metabolic pathways** and 43.25% of energy expenditure pathways, leading to lower amino acid, sugar, and protein content in grains (Wu et al., 2022, *Journal of Hazardous Materials*, 422, 126834, DOI: 10.1016/j.jhazmat.2021.126834). ScienceDirect PubMed Rice plants grown in microplastic-contaminated soil produced **35.45% empty shells**, and peanut seeds weighed 3.45% less. PubMed Central In lettuce, 0.1% microfiber exposure altered 14 of 46 identified metabolites, while 0.2% changed 17 metabolites (Zeb et al., 2022). Frontiers Biodegradable PBAT microplastics decreased chlorophyll concentrations in lettuce, ScienceDirect indicating that substitution of conventional with biodegradable plastics does not eliminate phytotoxicity risks. ScienceDirect

A 2025 field study on the California Central Coast (Tiwari and Sistla, 2024, *PNAS Nexus*, 3(10), pgae433, DOI: 10.1093/pnasnexus/pgae433) found that **all sampled farms exhibited plastic contamination** despite adherence to best management practices, and that macroplastic contamination negatively correlated with soil moisture, microbial activity, available phosphate,

and soil carbon pool size — with effects detectable at **less than 10% of contamination levels** previously thought necessary to degrade soil function. Oxford Academic

3.5 Crop vulnerability varies by growth habit and symbiotic dependencies

Different crop types exhibit markedly different susceptibilities to microplastic contamination. Emerging evidence suggests that **monocots (grasses, cereals) may be less sensitive than dicots** Frontiers (2025, *Frontiers in Plant Science*), though cereal grains still show significant nutrient profile changes at realistic exposure levels. Rice exhibits subspecies-dependent responses: PS microplastic exposure decreased head rice yield by **10.62% in variety Y900** but increased it by 6.35% in XS123 PubMed (Wu et al., 2022), highlighting the complexity of predicting impacts across cultivar diversity.

Legumes face particular vulnerability because of their dependence on nitrogen-fixing rhizobia and arbuscular mycorrhizal associations, both of which are disrupted by microplastic-induced changes in soil microbial communities. Frontiers Biodegradable PLA-PBAT at 1.5% severely affected common bean (*Phaseolus vulgaris*) growth (Meng et al., 2021), ScienceDirect while conventional LDPE at 1.0% actually increased leaf area ScienceDirect — demonstrating that polymer type matters as much as concentration. **Root crops** face direct physical exposure: microplastics adhere to and penetrate root surfaces, with carrot and radish showing impaired mineral translocation to edible portions. Frontiers Springer Polystyrene combined with arsenic caused oxidative bursts in carrot tissues, deteriorating quality (Dong et al., 2021). Memberclicks **Leafy vegetables** exhibit sensitive physiological and biochemical indicators, ScienceDirect and a breakthrough 2025 *Nature* study demonstrated that atmospheric microplastics are directly **absorbed through leaf stomata**, with open-air vegetables containing **10–100 times higher** microplastic concentrations than greenhouse-grown equivalents Scientific American (Li, Zhang, Xu et al., 2025, *Nature*, 641, 666–673, DOI: 10.1038/s41586-025-08831-4).

3.6 The Trojan horse effect amplifies heavy metal and organic pollutant toxicity

Microplastics serve as vectors for adsorbed contaminants, PubMed creating synergistic toxicity pathways that exceed the effects of either pollutant alone. MDPI +3 A meta-analysis of 57 studies found that microplastics **significantly increased soil cadmium bioavailability by 6.9%** and cadmium accumulation in plant shoots by 9.3%, with PE displaying a 29.4% stronger promoting effect among polymer types ScienceDirect (*Journal of Hazardous Materials*, 476, 135028, DOI: 10.1016/j.jhazmat.2024.135028). Combined microplastic and heavy metal exposure decreased plant root biomass and superoxide dismutase enzyme activity more than either pollutant alone, with **oxidative stress** identified as the common synergistic mechanism. Cas Cas

The Trojan horse pathway operates through physisorption: microplastics adsorb pollutants via Freundlich-model interactions, with aged and weathered particles showing enhanced adsorption capacity Aims Press (Carriera et al., 2023, *AIMS Environmental Science*, 10, 732–742, DOI: 10.3934/environsci.2023040). Sub-micron particles can penetrate root cells

Wiley Online Library carrying adsorbed heavy metals into internal plant tissues, constituting direct contaminant delivery to edible crop portions. MDPI PubMed Central In lettuce, microplastic presence significantly increased lead and cadmium concentrations while simultaneously decreasing copper accumulation, indicating complex element-specific interactions MDPI (2025, *Molecules*, DOI: 10.3390/molecules30112370). Phthalate leaching from plastic mulch films has produced concentrations in Chinese greenhouse vegetables (5.84 mg/kg in mustard, 3.62 mg/kg in celery) that exceed ambient soil levels. FAO Open Knowledge

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4. Detection methodologies remain fragmented and insufficient

4.1 No standardized protocol exists for soil microplastic analysis

The absence of standardized analytical methods for microplastic detection in soils represents a fundamental barrier to both scientific progress and regulatory action. Frontiers ScienceDirect Each step of the analytical pipeline — sampling, extraction, purification, identification, and quantification — lacks consensus protocols, producing results that vary by orders of magnitude across laboratories and rendering cross-study comparisons unreliable.

Fourier Transform Infrared (FTIR) spectroscopy and **micro-FTIR** identify polymers through characteristic infrared absorption fingerprints, detecting particles down to approximately 10–20 μm. However, particle-based micro-FTIR point mapping requires approximately 7.5 minutes per particle due to stage refocusing, making large-sample analysis prohibitively slow ACS Publications (Primpke et al., 2020, *Analytical and Bioanalytical Chemistry*, 412(30), 8283–8298, DOI: 10.1007/s00216-020-02979-w). **Raman spectroscopy** offers superior spatial resolution (~1 μm) but is highly susceptible to fluorescence interference from organic matter-rich soils. A new high-throughput screening Raman platform achieves measurements more than 300 times faster than particle-based micro-FTIR ACS Publications (2025, *ACS Omega*, DOI: 10.1021/acsomega.5c04522), but remains limited by matrix effects. **Pyrolysis-GC/MS** provides mass-based quantification with high sensitivity but destroys samples and cannot provide particle counts, sizes, or morphological information. Critically, PE overestimation from diagenetically altered organic matter has been documented in agricultural soils ACS Publications (Tanner et al., 2024, *Environmental Science & Technology*, 58(29), 13047–13055, DOI: 10.1021/acs.est.3c10101), raising concerns about the accuracy of mass-based polymer quantification. **Nile Red fluorescence staining** achieves recoveries of 80–90% for particles of 500–1,000 μm but lacks polymer-type specificity, and detection decreases sharply below 150 μm.

4.2 Inter-laboratory variation undermines data comparability

A major 2025 inter-laboratory comparison under the Versailles Project on Advanced Materials and Standards (VAMAS), involving **84 laboratories worldwide**, tested ISO-approved methods on reference materials PubMed and found relative standard deviations of 23% across

laboratories (*Analytical Chemistry*, 2025, DOI: 10.1021/acs.analchem.4c05403). Different spectral interpretation software packages contributed substantially to variability, with one commercial platform tending to overestimate particle counts. ACS Publications An earlier inter-laboratory comparison involving 17 laboratories from 8 countries showed "highly limited" comparability ResearchGate for particles in the 8–140 μm range, with underestimation more common than overestimation.

Extraction and separation procedures compound the variability. The most common density separation medium, sodium chloride (1.2 g/cm^3), cannot recover high-density polymers such as PVC (1.3–1.45 g/cm^3) and PET (1.38 g/cm^3). Zinc chloride (1.6–1.8 g/cm^3) recovers all polymers but is expensive and environmentally hazardous. Organic matter digestion methods — Fenton's reagent, enzymatic digestion, and alkaline hydrolysis — vary in effectiveness and may damage biodegradable polymers. Laboratory airborne contamination from indoor environments and equipment represents an additional, often uncontrolled source of procedural error. Frontiers The consequence is that without serial innovation designs isolating analytical from sample-preparation variability (2025, *Environmental Science & Technology*, DOI: 10.1021/acs.est.4c09427), measurement uncertainty remains prohibitively high for regulatory applications.

4.3 Emerging technologies offer promising but unproven solutions

Several emerging methodologies show potential for overcoming current limitations. **Laser Direct Infrared (LDIR) imaging** ltrcweb uses quantum cascade lasers for rapid scanning, analyzing up to 10^3 particles in 1–2 hours with detection down to 20 μm and 80–100% recovery for particles of 200–500 μm. Its performance in the 2025 VAMAS inter-laboratory comparison showed better agreement than micro-Raman or micro-FTIR. **Machine learning approaches** have achieved classification accuracies exceeding 95% for PE, PP, and PVC identification from Raman spectra using one-dimensional convolutional neural networks (Xu et al., 2023), while automated support vector classification models overlaid with thermogravimetric analysis data enable multi-polymer identification (2025, *Analytical Chemistry*, DOI: 10.1021/acs.analchem.4c06775).

Hyperspectral imaging coupled with machine learning can detect PE and polyamide in soil at concentrations as low as 1.6% using InGaAs and MCT cameras (Chen et al., 2024, *Journal of Hazardous Materials*, 471, 134346). The **FAO-IAEA** jointly launched a five-year Coordinated Research Project in 2023 to develop **isotopic tracing techniques** for tracking microplastic transport, degradation, and interaction with soil nutrients in agricultural systems. FAO Open Knowledge FAO Surface-enhanced Raman spectroscopy (SERS) holds promise for detecting sub-micron particles (Xu et al., 2020, *Environmental Science & Technology*, 54(24), 15594–15603), while aptamer-based biosensors are introducing molecular-level selectivity for specific polymers. Nevertheless, nanoplastics (< 1 μm) PubMed Central remain largely undetectable by all current routine methods, Frontiers representing a critical blind spot given their demonstrated ability to penetrate plant cell membranes and accumulate in edible tissues.

5. Regulatory frameworks are absent where they are most needed

5.1 No country has established soil quality standards for microplastics

Despite decades of accumulating contamination and a growing body of evidence documenting ecological and agricultural harm, **no country in the world has established regulatory limits for microplastic concentrations in agricultural soils**. (ltrcweb) This regulatory vacuum persists across all major agricultural regions and represents a fundamental failure to translate emerging science into protective policy.

In the **European Union**, the Sewage Sludge Directive (86/278/EEC), enacted in 1986, limits only six heavy metals and has not been substantively updated to address microplastics, pharmaceuticals, PFAS, or other emerging contaminants. A 2023 Joint Research Centre feasibility study acknowledged that current risk assessments are inadequate (Egle et al., 2023, DOI: 10.2760/305263). The EU has taken the most ambitious regulatory steps globally through its **REACH restriction on intentionally added microplastics** (EU 2023/2055), adopted in September 2023. This regulation prohibits the sale of products containing greater than 0.01% microplastics by weight, with agricultural and horticultural products subject to restrictions by October 2028. Agricultural polymers in fertilizers, seed coatings, and pesticides must demonstrate **90% degradation in soil within 48 months**. The revised **Urban Wastewater Treatment Directive** (EU 2024/3019), which entered force in January 2025, requires systematic monitoring of microplastics in wastewater treatment plant inlets and outlets, and mandates microplastic testing of sludge intended for agricultural use, with measurement methodologies to be established by July 2027.

In the **United States**, the EPA's biosolids regulations (40 CFR Part 503) do not mention microplastics. The Microbead-Free Waters Act of 2015 addresses only rinse-off cosmetics. A comprehensive 2021 survey by the Interstate Technology and Regulatory Council found that only 4 of 26 surveyed states had conducted any microplastic sampling, with **no established regulatory criteria or standards** for microplastics in any environmental medium. The growing PFAS crisis in biosolids — with the EPA's January 2025 draft risk assessment finding PFAS levels as low as **1 part per billion** potentially harmful to farmers — is accelerating regulatory attention to biosolid contaminants more broadly, with Maine becoming the first state to ban biosolid land application and over 10 states now actively regulating PFAS in land-applied materials.

China has focused primarily on mulch film management, setting a minimum plastic mulch thickness of 0.01 mm (national standard GB 13735-2017) and enacting Agricultural Film Management Measures in 2020 to regulate production, sale, use, and recycling. The 14th Five-Year Plan established a target of **85% agricultural film collection rate by 2025**. However,

enforcement remains uneven, and no concentration limits exist for microplastics in Chinese agricultural soils despite the nation having the most extensive contamination data globally.

5.2 The science-policy gap widens as evidence accumulates

The disconnect between scientific evidence and regulatory response is widening. The FAO's August 2025 landmark study, synthesizing over 2,000 peer-reviewed papers, concluded that soil plastic pollution is "**often irreversible**" and that adverse effects on soil properties, microbial life, and crop yields are observable even at concentrations as low as 0.001% by weight. The report stated: "Evidence suggests that after high concentration of plastics in soils reached during multiannual use, the initial benefits of plastic use for production reverse and result in decreased crop yields, jeopardizing food security." A 2025 risk assessment published in *Environmental Science & Technology* (DOI: 10.1021/acs.est.5c08026) found that under worst-case microplastic scenarios in sludge-amended soils, **39% of soil species would be affected**, and even realistic scenarios affected 15–18% of species — failing to protect 95% of soil biota, the conventional benchmark for environmental protection.

Current food safety standards, including the Codex Alimentarius and national food safety regulations, do not address microplastics in crops or food products. The FAO's 2022 food safety review (*Microplastics in Food Commodities*, Garrido Gamarro and Costanzo, Food Safety and Quality Series No. 18) acknowledged "generally low" reported levels in food but flagged critical gaps in exposure data, toxicity knowledge, and analytical standardization. The absence of reference methods makes enforceable standards impossible to establish — a circular problem that demands coordinated international investment in methodology development as a prerequisite for regulation.

5.3 Food security implications are most severe for nations dependent on biosolids and plastic mulch

The food security implications of agricultural microplastic contamination are geographically concentrated but globally consequential. **China's** northwest regions — Xinjiang, Gansu, and Inner Mongolia — are the most severely affected, with cotton, vegetable, and grain production critically dependent on plastic mulch in arid conditions where residue levels reach 259 kg/ha. Europe applies **31,000–43,000 tonnes of microplastics annually** via sewage sludge to farmland, with Ireland, Spain, the UK, and France applying the highest proportions. The United States land-applies approximately 60% of its biosolids, potentially affecting **nearly 70 million acres** according to Environmental Working Group estimates.

Developing nations face disproportionate risks. Low- and middle-income countries often lack analytical capacity for microplastic monitoring, depend on imported agricultural inputs containing microplastics, and have limited ability to implement alternative soil amendment strategies. Indonesia has the highest estimated global per capita microplastic dietary intake at approximately 15 grams per month. African nations face compounded challenges from rapid urbanization, informal waste management systems, and expanding agricultural activity — yet

are the least studied region globally. The PNAS 2025 meta-analysis warning of potential 4–14% global crop losses specifically notes that the resulting food insecurity burden would fall most heavily on regions already experiencing food deficit, with the potential to increase people at risk of starvation by approximately 400 million.

The intersection with other stressors magnifies the concern. Climate change is expanding the geographic range where plastic mulch is considered agronomically necessary for moisture conservation, while simultaneously increasing biosolid production as wastewater treatment capacity grows in developing countries. FAO data indicate that agricultural plastic demand is projected to increase **50% by 2030** (from 6.1 to 9.5 million tonnes), compounding contamination even as global food demand rises with population growth toward 9.7 billion by 2050.

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6. Toward evidence-based policy: what must change

The FAO's 2021 landmark report, *Assessment of Agricultural Plastics and Their Sustainability: A Call for Action*, established that agricultural value chains use **12.5 million tonnes of plastics** annually for production and 37.3 million tonnes for food packaging, and that soils contain more microplastics than oceans. In February 2025, the FAO published a Provisional Voluntary Code of Conduct on the Sustainable Use and Management of Plastics in Agriculture, endorsed by the FAO Council. The ongoing UN Intergovernmental Negotiating Committee (INC) process for a Global Plastics Treaty, though it failed to reach agreement in Busan in late 2024, represents the most promising vehicle for establishing binding international frameworks. The FAO has advocated for the inclusion of agriculture as a high-impact sector in treaty provisions.

Five priority actions emerge from this synthesis of the evidence:

- **Establish precautionary maximum allowable concentrations** for microplastics in agricultural soils and biosolids, using tiered approaches based on available ecotoxicological data, with revision as standardized methods mature
- **Develop and mandate internationally standardized analytical protocols** through ISO Technical Committee coordination, including certified reference materials for soil-matrix-specific analysis, enabling comparable monitoring data across jurisdictions
- **Invest in alternatives to plastic-dependent agricultural practices**, including biodegradable mulch films that meet verified degradation standards, enhanced biosolid treatment technologies for microplastic removal, and nutrient recovery through phosphorus extraction from incineration ash
- **Implement systematic monitoring programs** in developing nations, with capacity-building support, to address the critical data gap in regions where food security is most fragile

- **Apply the One Health framework** recognizing the interconnection of soil, plant, animal, and human health, integrating microplastic risk assessment into existing food safety, soil quality, and water management regulatory architectures

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7. Conclusion: an irreversible contamination trajectory demanding precautionary action

This synthesis reveals a contamination problem that is far larger, more persistent, and more consequential than current policy frameworks acknowledge. Agricultural soils globally contain median microplastic concentrations of approximately **1,000–3,000 particles per kilogram**, with heavily amended soils exceeding 20,000 particles per kilogram and concentrations increasing with each application cycle. These particles persist for decades without meaningful degradation, accumulate in the root zone where crop nutrient uptake occurs, and progressively fragment into nanoplastics capable of penetrating plant tissues directly.

The nutritional security dimension of this contamination has been largely overlooked. Microplastic exposure reduces essential micronutrient concentrations in cereal grains — **zinc, calcium, and manganese decline by 10–17%** in rice — while impairing photosynthesis by 7–12% and reducing overall productivity by up to 14.5%. The mechanisms are now well characterized: altered soil structure, disrupted microbial communities (including mycorrhizal fungi essential for nutrient transfer), impaired root function, and competitive sorption of nutrients by plastic particles. The Trojan horse effect, whereby microplastics increase heavy metal bioavailability and facilitate contaminant delivery into plant tissues, adds a further dimension of concern for food safety.

Perhaps the most important insight from this review is the **irreversibility** of the contamination trajectory. Unlike many agricultural soil contaminants that can be remediated or that degrade over time, microplastics accumulate persistently and continuously. The 25-year sludge study showing unchanged microplastic levels after more than two decades, combined with the ongoing fragmentation of existing macroplastics into new microplastic particles, means that current contamination levels represent a floor, not a ceiling. Every year of continued unregulated biosolid application and plastic mulch use adds a permanent increment to the soil microplastic burden.

The regulatory response has been profoundly inadequate. Not a single jurisdiction has established soil quality standards for microplastics. The EU's recent steps — the REACH restriction on intentionally added microplastics and the revised wastewater monitoring requirements — are necessary but insufficient, as they address inputs rather than the accumulated legacy contamination already present in soils. The forthcoming Global Plastics Treaty represents the most consequential policy opportunity, but only if agricultural sector provisions are included with binding commitments to monitoring, reduction, and remediation.

The scientific community has delivered a clear message through multiple lines of converging evidence: agricultural microplastic contamination is degrading crop nutrition, threatening soil function, and undermining the productive capacity of the world's farmland at a time when that capacity has never been more critically needed.